

COMBINED RADIATION AND CONDUCTION HEAT TRANSFER SOLVER ON HYBRID CPU-GPU ARCHITECTURES

Faizan P. Siddiqui, Kaan Meneksedag, Altug M. Basol, M. Pinar Mengüç

Mechanical Engineering Department,
Ozyegin University, Istanbul, Turkey

ABSTRACT: Heat transfer phenomena combining conduction and thermal radiation are present in multiple industrial processes. In this work, a numerical approach for solving combined mode heat transfer problems is discussed. The thermal radiation part is solved using Monte Carlo ray tracing (MCRT) method on the graphic processors. Performance tests on a surface-to-surface heat exchange scenario have shown a speed-up advantage of 90x with the use of a high-end gaming GPU in comparison to the four-core of a desktop CPU. The accelerated thermal radiation solver is integrated into the existing conduction heat transfer solver in OpenFoam and the hybrid solver is tested on three-dimensional heat transfer problems involving non-scattering but absorbing and emitting media. Performance tests on the solver shows the dominance of the thermal radiation part of the problem in the computation cost. Further test on the performance of the thermal radiation solver on a three dimensional, participating medium scenario showed a rather weak effect of the mesh size or the optical thickness on the computational cost of the method.

INTRODUCTION:

Scenarios including both thermal radiation and conduction are present in a wide range of industrial applications and numerical solutions to these combined mode heat transfer problems are of considerable interest for the heat transfer community. Apart from scenarios where the medium is optically very thick, these two heat transfer modes are very different in nature. In presence of optically thick medium, thermal radiation can travel only a very limited distance before getting absorbed by the medium. In this regard, the direct heat exchange over a long distance is prevented by the thickness of the medium and thermal radiation can also be modeled as a diffusion like phenomena. In the P_1 approximation of the Spherical Harmonics Method (SHM), the radiative transfer equation (RTE) is converted into a single elliptic partial differential equation and is capable of accurately capturing the physics of the problem with a low computational cost. Menguc and Viskanta [1] have applied the higher order P_3 approximation of the method in a combustion system containing anisotropically scattering participating medium. Modest et. al. [2] have developed the simplified version of the higher order approximations of the Spherical Harmonics method, called SP_N , and applied them in media with optically varying thicknesses.

On the other hand, in an optically thin medium thermal radiation can penetrate into the medium and affect the direct heat exchange over distant regions in the domain. The change in the physics of the problem with the reduction in the optical thickness of the medium makes the methods with the diffusion approximation invalid. Finite Volume Method [3] and Discrete Ordinate Methods [4] are much better suited for participating medium with low optical thickness. Sun et. al [5] have tested FVM both in accuracy and computational cost in comparison to Spherical Harmonics type methods in a 2D combined conduction-radiation problem. Numerical tests showed the advantage of the FVM over the SHM in terms of accuracy and computational cost in participating media with low and medium optical thicknesses. Selcuk and Kayakol [6] have pointed out the accuracy of the DOM in media with low optical thicknesses. Despite their superiority over the methods using the diffusion approximation in low

optical thickness, both of these methods are prone to discretization errors as discussed by Chai et. al [7] and Coelho [8]. Especially in scenarios involving complex geometries these errors may increase due to the relatively low mesh quality.

The lack of the discretization errors makes the Monte Carlo ray tracing (MCRT) method very suitable for the problems including optically thin media. It is also much better suited for applications involving complex geometries. However, especially in low optical thickness media, the method might require tracing very high number of rays to decrease the statistical errors to a reasonable level. In this regard, this method is mostly applied in providing the reference solution for the validation scenarios. Howell [9] has discussed the method and its advantages in detail.

MCRT has gained popularity over the years parallel to the drop in the cost of computational power and due to its the perfect suitability to today's multi and many-core computer architectures. Parallelization strategies for MCRT on the vector computers has been discussed by Brown and Martin [10] Farmer and Howell [11] have in depth analyzed different parallelization strategies of the method in shared and distributed memory computer architectures and they have achieved near ideal speed-up levels using SIMD parallelization paradigm.

Graphic processors units (GPU) are type of processors specifically designed for rendering applications. The intrinsically parallel nature of the rendering computations required the chip to develop very differently from the CPU in architecture. The dramatic increase in their computational power over the last years and the emergence of the high level programming models like CUDA or OpenCL spread the use of the GPU's in scientific computing field. Fang and Boas [12] have parallelized Monte Carlo photon migration method using CUDA and obtained speedup values above 300 using a low-cost graphic card over the single core of an Intel Xeon CPU. Takizawa et. al [13] have solved for the exchange factors between surfaces on the GPU and pointed out the speed advantage of the GPU's. Halverson [14] presents the solar radiation modeling of an urban area using GPU and Nvidia's Optix engine, He et al. [15] solved radiation in combustion and propulsion applications utilizing graphic processors. Siddiqui et al. [16] used GPU for solving surface to surface radiative heat transfer problems involving complex three-dimensional geometries.

In this study, a numerical approach for solving heat transfer problems combining conduction and radiation is developed. The details of the backward MCRT based thermal radiation solver is discussed and the gain in the computational speed by GPU acceleration is evaluated on a surface-to-surface radiation exchange problem. The developed thermal radiation solver is integrated into the OpenFoam framework to solve for the coupled radiation-conduction problem. The performance of the hybrid (CPU-GPU) solver in terms of computational cost is explored on a 3D validation scenario with different mesh resolutions and optical thicknesses.

NUMERICAL METHOD

The present study is focused in the solution of coupled radiation conduction problem for three dimensional complex geometries. The energy equation combining the conduction and radiation heat transfer modes is given in Eq. 1. First term on the right hand side represents the heat transfer with the conduction, while the second term represents the divergence of the radiative flux at any arbitrary point $\mathbf{r}$.

$$\rho c \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) - \nabla \cdot \mathbf{q}_r \quad (1)$$

The effect of thermal radiation is included as explicit source term in the energy equation and it is calculated as described in Eq. (2).

$$-\nabla \cdot \mathbf{q}_r(\mathbf{r}) = \kappa \cdot \left(\int_{4\pi} I(\mathbf{r}, \hat{\mathbf{s}}_i) d\Omega_i - 4\pi \cdot I_b(\mathbf{r}) \right) \quad (2)$$

The evaluation of the integral term on the right hand side requires the solution of the Radiative Transfer Equation (RTE), an integro-differential equation with six independent parameters [17] In differential form the RTE can be expressed as:

$$\frac{\partial I_\lambda(S, \Omega)}{\partial S} = \kappa_\lambda I_{\lambda b}(S, \Omega) - \kappa_\lambda I_\lambda(S, \Omega) - \sigma_{s,\lambda} I_\lambda(S, \Omega) + \frac{1}{4\pi} \int_{\Omega_i=4\pi} \sigma_{s,\lambda} I_\lambda(S, \Omega) \Phi_\lambda(\Omega_i, \Omega) d\Omega_i \quad (3)$$

where the term on the left hand side of the equation expresses the time rate of change in the spectral intensity per unit projected area and the terms on the right hand side dictate the effects of emission, absorption and scattering onto the intensity of the light.

A collision-based backward Monte Carlo ray tracing method [Modest] was used for the solution of the RTE. In this method the rays are traced in backward direction from the target to the source of the emission. The random numbers in the method are generated using Tiny Encryption Algorithm [18]. The discussed method is implemented on the GPU with the use of the Optix Ray Tracing Engine [19], a ray tracing library developed by Nvidia.

The divergence of the radiative heat flux is first computed for each cell on the GPU and after that they are transferred to the CPU where they are considered as a source term in the energy equation. The energy equation with the conduction heat transfer is solved on the CPU using OpenFoam [20], an open source library for the solution of partial differential equations. The updated temperature values are sent back to the GPU to recalculate the divergence of the radiative heat fluxes. The iterations are carried on until a certain residual value is reached. The flowchart of the program is given in Figure 1.

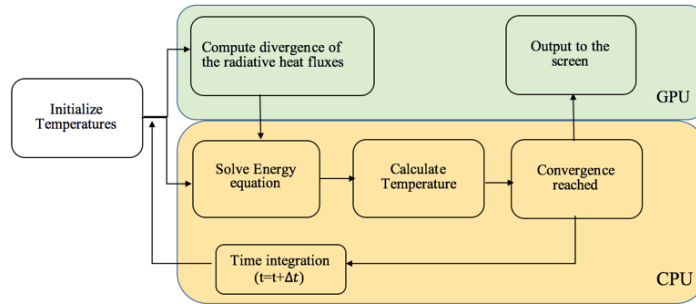


Figure 1 Flowchart of the hybrid solver

Validation of the Numerical Method

Thermal radiation in absorbing-emitting media without conduction heat transfer

The developed MCRT based thermal radiation solver is tested on a validation scenario discussed by Hsu and Farmer [21]. The same test problem involving 3d participating medium is considered previously by Henson [22] and Guo and Maruyama [23] as well. The test problem consists of a unit cube enclosing an isothermal, absorbing, emitting medium. All the surfaces of the enclosure are cold and black such that there is no emission or reflection from the boundaries. The extinction coefficient inside the cube is defined by the relation:

$$\beta(x, y, z) = 0.9(1 - 2|x|)(1 - 2|y|)(1 - 2|z|) \quad (4)$$

The coordinate origin lies at the center of the cube extending by 0.5 in length in every direction. The participating medium has unity black body emissive power. First, the domain is sub-divided into 9

uniform and Cartesian cells in every direction. The centers of the cells are located at $(x, y, z) = (0, \pm 1/9, \pm 2/9, \pm 3/9, \pm 4/9)$. For every cell, the extinction coefficient is evaluated from equation 4 and assumed to be constant within the cell. Following the collision based Monte Carlo approach, the divergence of radiative flux for every cell in the domain is calculated. In total, 5 million rays are used in the calculation. Table 1 shows the divergence of the radiative flux of the cells at the centerline of the cube calculated with the present method along with the other studies from the literature. As shown in Table 1 the results of the present tool are in agreement with the results from the literature.

Table 1: Divergence of the radiative heat flux at $(x,0,0)$

X	MCRT[Present]	Uncertainty	MCRT[21]	MCRT[22]	DT	YIX
$\pm 4/9$	0723284.	0.000257	0.72910	0.72336	0.72860	0.72219
$\pm 3/9$	1.376448	0.000515	1.38739	1.37701	1.38099	1.37208
$\pm 2/9$	1.969357	0.000851	1.98360	1.96893	1.96458	1.95652
$\pm 1/9$	2.519262	0.001158	2.53635	2.51700	2.52182	2.49628
0	3.07424	0001449	3.09813	3.08762	3.0844	3.03661

Thermal radiation in absorbing-emitting media with Conduction Heat Transfer

Next, the hybrid solver is tested on a validation scenario discussed by Viskanta [24] . The problem consists of conduction and radiation heat transfer between infinitely long parallel plates. Important parameters for the validation scenario are provided in Table 2.

Table 2: Parameters for the validation scenario

Parameters		Value
κ	$[\text{m}^{-1}]$	328.08
k	$[\text{W/mK}]$	0.093
T_{Top}	$[\text{K}]$	1922
T_{Bottom}	$[\text{K}]$	1088

The one-dimensional problem is solved using a three dimensional geometry where the plates are set to a length $L = 0.66$ m in both lateral directions and the plates are placed at a distance of $d = 0.003028$ m from each other. The distance between the plates is divided into 39 uniform cell layers composed of 39×39 cells in lateral directions. The total mesh consists of $39 \times 39 \times 39$ cells.

To mimic the one-dimensionality of the scenario, the rays are only released from the cells located at the centerline of the domain and traced in all the directions within the domain. With the provided absorption coefficient, the probability of a ray released from the centerline of the domain reaching the side boundaries is one part in a billion. It is concluded that having the domain truncated in lateral direction does not have a considerable effect on the solution. After calculating the divergence of the radiative flux for the cells along the centerline, the calculated divergence values are shared with all the cells (39×39) across the same cell layer. Finally, the energy equation is solved on all the cells with the provided divergence of heat fluxes from the thermal radiation solver. Adiabatic boundary conditions are imposed at the sides of the domain to comply with the one-dimensionality of the validation scenario.

The convergence of the solution is assessed by monitoring the overall convergence level for the energy equation. With the divergence of heat flux term received from the GPU, the energy equation is solved until the residual drops below $1\text{e-}12$. However, the recalculation of the divergence of the heat flux on the GPU with the updated temperature levels, rises back the residual in the energy equation. Figure 2 shows the initial residual levels obtained with the updated source term received from the GPU before solving for the energy equation. As shown, the residual level first drops and after certain number of

iterations it stagnates at a level depending on the ray count used in the calculation of the divergence of heat fluxes. Increasing the ray count, reduces the randomness in the divergence of the heat flux values and that leads to a reduced initial residual in the energy equation. Increasing the ray count by an order of magnitude reduces the residual in the energy equation by a factor of only 3. A residual level lower than $1e-4$ requires tracing 100,000 rays per cell indicating the computational cost of the MCRT method. Figure 3a shows the analytical and the numerical solution obtained releasing 10,000 rays per cell. Furthermore, the error between the analytical and the numerical solution with different ray counts is plotted in Figure 3b.

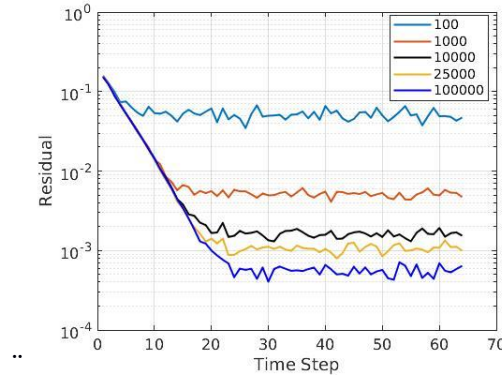


Figure 2 The convergence behavior of the combined heat transfer problems with varying ray counts used in the solution of the thermal radiation part of the problem

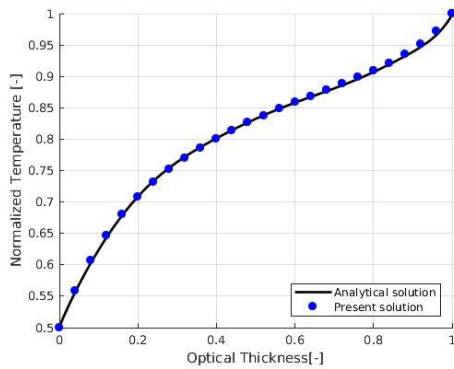


Figure 3a Analytical vs. Numerical Solution

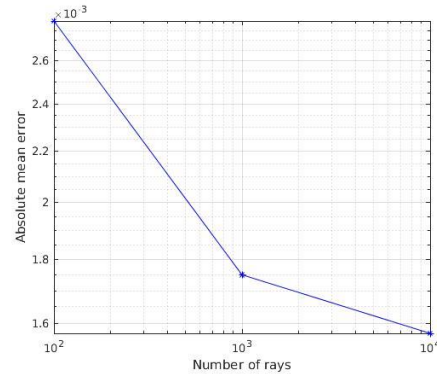


Figure 3b Error vs. Number of rays traced

RESULTS AND DISCUSSION

The computational cost of the MCRT method is dominated by the ray tracing calculations. The speed-up achieved by the use of GPU was analyzed on a surface-to-surface heat exchange calculation between sphere and rectangular plate [25]. The exchange factor calculations are conducted using MCRT method. The implementation is carried out in a way that enables the same code to run both on the CPU and on the GPU. The computation times required for tracing a certain number of rays on the four-cores of an Intel desktop CPU and on a high end Nvidia gaming GPU are compared in Figure 4. As shown, the present GPU is capable of tracing around 250 million rays per second on this surface-to-surface heat exchange problem. This comparison shows that a single high-end gaming GPU can accelerate the calculations by a factor of 90 in comparison to a four-core Intel desktop CPU.

The performance of the solver is then evaluated using the same combined conduction-radiation scenario used in validating the solver. The parameters for the test case are as in Table 2. With the chosen absorption coefficient, the optical thickness across the plates is set $\tau = 1$. This represents a scenario with a medium optical thickness. The computational domain is divided into $39 \times 39 \times 39$ uniform, structured

hexahedral cells. First, the computational cost of running the scenario completely three-dimensionally is evaluated. The computational cost of the problem heavily depends on the number of rays traced in the MCRT method. The number of rays traced per cell is fixed to 25,000. As shown in Figure 2, this ray count per cell results in a residual level of $1.0e-3$ in the energy equation. In total, 1.5 billion rays are traced per time step. The energy equation with the conduction heat transfer and the source term taken from the MCRT calculation is solved on a single core of the available CPU and the MCRT calculation is carried out on the GPU. Table 3 shows the computation time required to solve the thermal radiation and conduction heat transfer separately. The results indicate the dominance of the thermal radiation calculation in the computational cost of the problem. Solving for the thermal radiation using 1.5 billion rays takes about 85 seconds, meaning that only around 18 million rays can be traced in a second. This indicates a major reduction in the performance of the GPU in participating medium problem in comparison to the surface-to-surface exchange factor calculations discussed previously. The drop in the number of rays traced is mainly due to the multiple intersections between each ray traced and the cells in their trajectories. On the other hand, in the surface-to-surface heat exchange calculation each ray intersects the cell surface at most once.

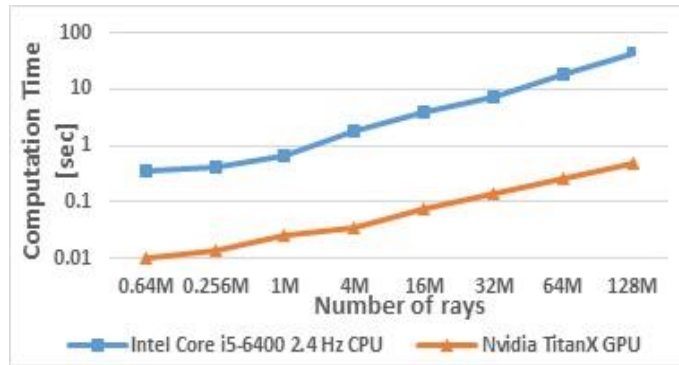


Figure 4. Computation time comparison between CPU and GPU

Table 3: Computation time for the solution of coupled radiation conduction problems

	Computation Time [s]
Conduction and Energy equation (on CPU)	0.096
Thermal Radiation (on GPU)	85.4
Total	85.5

Since the overall computational cost is completely dominated by the calculation of the thermal radiation, further performance tests are conducted regarding thermal radiation solver. As a next scenario, the effect of the variation in the optical thickness of the medium in the computational cost of the method is studied. Decreasing the optical thickness increases the penetration of the rays into the medium and that increases the number of ray-cell face intersections and thereby the computation cost. Moreover, decreasing the optical thickness, requires an increase in the ray count to keep the same uncertainty level in the results. Figure 5a shows the number of rays required to reach the same residual level ($1.0e-3$) for all the different cases. Scenario with a low optical thickness $\tau = 0.1$ requires tracing 100,000 rays per cell, a factor of 4 increase in the ray count. In Figure 5b, the computation time of running the MCRT calculations for different optical thicknesses are compared. The mesh size is fixed to $39 \times 39 \times 39$ for all the cases. The tests are conducted by keeping either the ray count constant or the residual level. Keeping the same number of rays, a factor of 10 change in the optical thickness results into a factor of 2 change in the computation time. The dependency of the computation time on the optical thickness is slightly higher for a constant residual level.

Finally, the effect of the mesh size on the computational cost is studied. For this study, the overall ray

count is fixed for all the different mesh sizes and optical thickness of the medium is set to $\tau = 1$. In addition to the existing mesh size $39 \times 39 \times 39$, two additional meshes are generated: one finer ($79 \times 79 \times 79$) and one coarser ($19 \times 19 \times 19$). Figure 6a shows the computation time required to solve the thermal radiation calculation with the three sets of meshes. Results indicate a factor of 2x increase in the computation time with an 8x increase in the mesh size. The increase in the computation cost with the finer mesh resolution is due to the increase in the mesh-cell intersections that is only doubling by a one-level refinement in the mesh in every direction. With the current implementation, the fine mesh consisting of $79 \times 79 \times 79$ cells requires about 250 seconds on the GPU, still two orders of magnitude higher than the computation time required to solve the conduction heat transfer on the CPU. (shown in Figure 6b)

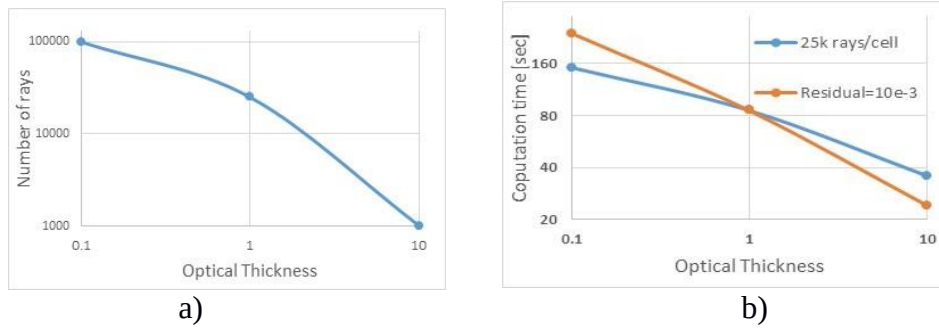


Figure 5 Number of rays required to reach to the residual level 1.0×10^{-3} for different optical thicknesses (a), the computation time vs. optical thickness (b)

CONCLUSION:

The performance benefit of the GPU in MCRT calculations is quantified on a surface-to-surface heat exchange problem.

- The high end gaming GPU can trace around 250 million rays per second in a surface-to-surface heat exchange scenario.
- The GPU shows a performance benefit of around 90x in comparison to a four-core desktop CPU.

A hybrid solver for the solution of combined radiation and conduction heat transfer problems is developed and validated on scenarios involving non-scattering but absorbing and emitting media. The computational cost of the method is dominated by the thermal radiation calculation even though it is accelerated on the GPU. Performance tests on the thermal radiation part of the solver shows that

- the current implementation running on the GPU is capable of tracing of 18 million rays per second, a much lower ray count per second than with the surface-to-surface scenario.
- The optical thickness of the medium is found to have a rather weak effect on the computational cost of the method. A factor of 10 change in the optical thickness results in a change of around a factor of 2 in the computational cost of the thermal radiation solver.
- The computational cost of the method got only weakly affected by the overall mesh size. An increase in the mesh size by a factor of 8 resulted in an increase of a factor 2 in the computation cost of the method.

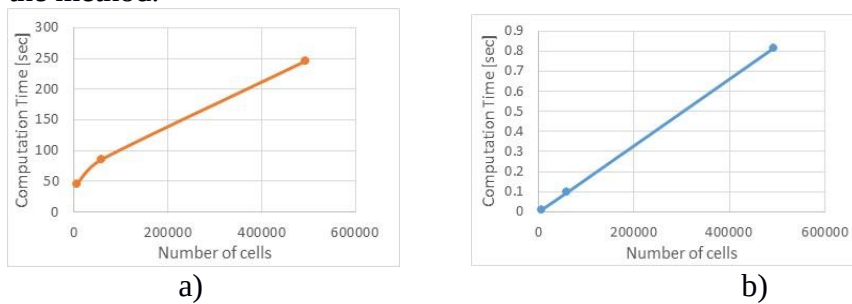


Figure 6. Computation time vs number of cells for solving Radiative heat flux (a), Computation time vs number of cells for the solution of energy equation (b)

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